

Packet Tracer – Network Troubleshooting Complex Scenario

Objectives

Part 1: Build a Multi-Network Enterprise Topology

Part 2: Configure a Working Network

Part 3: Introduce Multiple Network Problems

Part 4: Troubleshoot VLAN Issues

Part 5: Troubleshoot ACL Issues

Part 6: Troubleshoot Routing Issues

Part 7: Restore Full Connectivity

Background / Scenario

You are a junior network administrator who has been called to troubleshoot a network outage.

The network was functioning correctly yesterday, but several configuration changes were made overnight.

Users report:

- HR PCs cannot reach the Sales network
- Sales PCs cannot reach the server
- Some devices cannot reach their default gateway
- Inter-VLAN communication has failed

Management wants all services restored as quickly as possible.

Your job is to identify and fix:

- VLAN configuration errors
- Trunking errors
- ACL misconfigurations
- Routing mistakes
- Incorrect IP addressing

Unlike previous labs, this network starts **broken on purpose**.

Instructions

Part 1: Build the Topology

Step 1: Add Devices

Add:

- 1 Router (2911)
- 2 Switches (2960)
- 1 Server
- 4 PCs

Rename:

- Router0 → R1
 - Switch0 → SW1
 - Switch1 → SW2
 - Server0 → SERVER
 - PC0 → HR-PC
 - PC1 → SALES-PC
 - PC2 → IT-PC
 - PC3 → GUEST-PC
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Step 2: Connect Devices

Use Copper Straight-Through cables.

Device	Interface	Device	Interface
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HR-PC	Fa0	SW1	Fa0/1
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Device	Interface	Device	Interface
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SALES-PC	Fa0	SW1	Fa0/2
SW1	Fa0/24	SW2	Fa0/24
IT-PC	Fa0	SW2	Fa0/1
GUEST-PC	Fa0	SW2	Fa0/2
SERVER	Fa0	SW2	Fa0/3
SW1	Fa0/23	R1	G0/0

Wait until all links turn green.

Part 2: Configure a Working Network

VLAN Plan

VLAN	Name	Network
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10	HR	192.168.10.0/24
20	SALES	192.168.20.0/24
30	IT	192.168.30.0/24
40	GUEST	192.168.40.0/24

Configure VLANs on Both Switches

enable

configure terminal

vlan 10

name HR

vlan 20

name SALES

```
vlan 30  
name IT
```

```
vlan 40  
name GUEST
```

```
end
```

Assign Access Ports

SW1

```
interface fa0/1  
switchport mode access  
switchport access vlan 10
```

```
interface fa0/2  
switchport mode access  
switchport access vlan 20
```

SW2

```
interface fa0/1  
switchport mode access  
switchport access vlan 30
```

```
interface fa0/2  
switchport mode access  
switchport access vlan 40
```

```
interface fa0/3  
switchport mode access  
switchport access vlan 30
```

Configure Trunk Link

SW1

```
interface fa0/23
switchport mode trunk
```

```
interface fa0/24
switchport mode trunk
```

SW2

```
interface fa0/24
switchport mode trunk
```

Configure Router-on-a-Stick

```
enable
configure terminal
```

```
interface g0/0
no shutdown
```

```
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

```
interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
```

```
interface g0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
```

```
interface g0/0.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
```

```
end
```

Configure PC Addressing

HR-PC

IP Address: 192.168.10.10
Subnet Mask: 255.255.255.0
Gateway: 192.168.10.1

SALES-PC

IP Address: 192.168.20.10
Subnet Mask: 255.255.255.0
Gateway: 192.168.20.1

IT-PC

IP Address: 192.168.30.10
Subnet Mask: 255.255.255.0
Gateway: 192.168.30.1

GUEST-PC

IP Address: 192.168.40.10
Subnet Mask: 255.255.255.0
Gateway: 192.168.40.1

SERVER

IP Address: 192.168.30.100
Subnet Mask: 255.255.255.0
Gateway: 192.168.30.1

Part 3: Verify the Working Network

From HR-PC:

ping 192.168.20.10
ping 192.168.30.100

Expected:

Reply received

At this stage the network should work.

Part 4: Break the Network

Now intentionally introduce problems.

Problem 1: Wrong VLAN Assignment

On SW1:

```
interface fa0/2  
switchport access vlan 10
```

Instead of VLAN 20.

Problem 2: Trunk Failure

On SW2:

```
interface fa0/24  
shutdown
```

Problem 3: Routing Error

On R1:

```
interface g0/0.30  
  
ip address 192.168.30.254 255.255.255.0
```

Instead of:

```
192.168.30.1
```

Problem 4: ACL Error

Create ACL:

```
access-list 100 deny ip any any
```

Apply:

```
interface g0/0.20
```

```
ip access-group 100 in
```

Part 5: Troubleshooting Tasks

The network is now broken.

You must identify all problems.

Check Interfaces

```
show ip interface brief
```

Look for:

administratively down

Check VLAN Assignments

```
show vlan brief
```

Look for:

Fa0/2 assigned to wrong VLAN

Check Trunks

```
show interfaces trunk
```

Look for:

No active trunk

Check Router Subinterfaces

```
show running-config
```

Look for:

Incorrect IP address

Check ACLs

show access-lists

Look for:

deny ip any any

Test Connectivity

Use:

ping

tracert

to identify where traffic stops.

Part 6: Fix the Network

Fix VLAN

interface fa0/2

switchport access vlan 20

Fix Trunk

interface fa0/24

no shutdown

Fix Routing

```
interface g0/0.30
```

```
ip address 192.168.30.1 255.255.255.0
```

Remove ACL

```
interface g0/0.20
```

```
no ip access-group 100 in
```

```
no access-list 100
```

Part 7: Verify Connectivity

Test HR to Sales

```
ping 192.168.20.10
```

Expected:

Reply received

Test HR to Server

```
ping 192.168.30.100
```

Expected:

Reply received

Test Guest to Server

```
ping 192.168.30.100
```

Expected:

Reply received

Verify Routing

show ip route

Expected:

Connected routes for all VLANs

Verify Trunks

show interfaces trunk

Expected:

Fa0/24 trunking

Verify VLANs

show vlan brief

Expected:

Correct VLAN membership

Troubleshooting Tips

If connectivity still fails:

✓ Check interface status

show ip interface brief

✓ Check VLAN assignments

show vlan brief

✓ Check trunk links

show interfaces trunk

✓ Check routing

show ip route

✓ Check ACLs

show access-lists

✓ Verify PC gateways

ipconfig

✓ Verify router subinterfaces

show run

What You Learned

- How to troubleshoot VLAN problems
 - How to troubleshoot trunk failures
 - How to identify ACL issues
 - How to verify routing configurations
 - How to use Cisco show commands effectively
 - How to systematically isolate network failures
 - How enterprise troubleshooting combines multiple networking technologies
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Challenge Extensions (Optional)

Challenge 1: Add DHCP Failure

Misconfigure:

ip helper-address

Troubleshoot DHCP requests.

Challenge 2: Add STP Failure

Block a trunk port accidentally and restore connectivity.

Challenge 3: Add EIGRP Misconfiguration

Configure incorrect AS numbers and identify why routes are missing.

Challenge 4: Add NAT Failure

Configure NAT incorrectly and troubleshoot Internet access.

Challenge 5: Simulation Mode Investigation

Switch Packet Tracer to:

Simulation Mode

Generate:

- ICMP
- ARP
- Routing traffic

Observe exactly where packets are dropped and identify the root cause before fixing the issue.